

MA 301

Test 3

Fall - 2014

Problem	points	out of
1		8
2		8
3		8
EC		2
Score		24

Name: SOLUTIONS

HR: _____

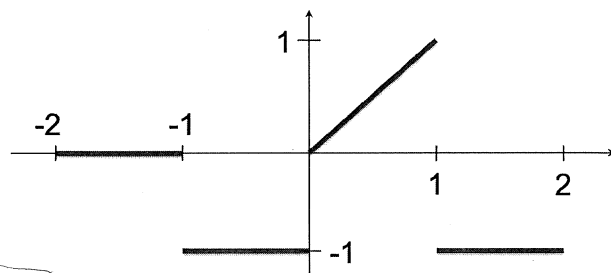
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READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.

$$d_0 = \frac{1}{4} \int_{-2}^2 f(x) dx = \frac{1}{4} \left(1 + \frac{1}{2} + 1 \right) = \frac{5}{8}$$



$$P = 4 \Rightarrow L = 2$$

$$d_n = \frac{1}{4} \int_{-2}^2 f(x) e^{-i\frac{\pi}{2}nx} dx = \frac{1}{4} \left(\underbrace{-\int_{-1}^0 e^{-i\frac{\pi}{2}nx} dx}_{(1)} + \underbrace{\int_0^1 x e^{-i\frac{\pi}{2}nx} dx}_{(2)} + \underbrace{\int_1^2 e^{-i\frac{\pi}{2}nx} dx}_{(3)} \right)$$

$$(1) = \left. \frac{2}{i\pi n} e^{-i\frac{\pi}{2}nx} \right|_{-1}^0 = \frac{2}{i\pi n} e^0 - \frac{2}{i\pi n} e^{i\frac{\pi}{2}n} = \frac{2}{i\pi n} - \frac{2}{i\pi n} i^n$$

$$(3) = \left. -\frac{2}{i\pi n} e^{-i\frac{\pi}{2}nx} \right|_1^2 = -\frac{2}{i\pi n} \left(e^{-i\pi n} - e^{-i\frac{\pi}{2}n} \right) = -\frac{2}{i\pi n} (-1)^n + \frac{2}{i\pi n} (-i)^n$$

$$(2) \quad \begin{array}{l} u = x \\ v = \frac{-2}{i\pi n} e^{-i\frac{\pi}{2}nx} \\ du = dx \\ dv = e^{-i\frac{\pi}{2}nx} dx \end{array}$$

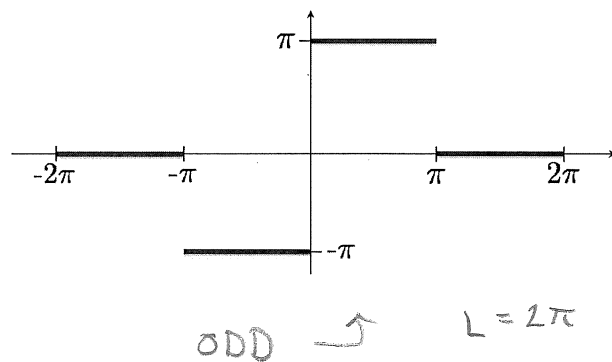
$$\begin{aligned} (2) &= \frac{-2x}{i\pi n} e^{-i\frac{\pi}{2}nx} + \frac{2}{i\pi n} \int_0^1 e^{-i\frac{\pi}{2}nx} dx \\ &= \frac{-2x}{i\pi n} e^{-i\frac{\pi}{2}nx} + \frac{4}{\pi^2 n^2} e^{-i\frac{\pi}{2}nx} \bigg|_0^1 \\ &= \frac{-2}{i\pi n} e^{-i\frac{\pi}{2}n} + \frac{4}{\pi^2 n^2} e^{-i\frac{\pi}{2}n} - \frac{4}{\pi^2 n^2} \\ &= \frac{-2}{i\pi n} (-i)^n + \frac{4}{\pi^2 n^2} (-i)^n - \frac{4}{\pi^2 n^2} \end{aligned}$$

So

$$\begin{aligned} d_n &= \frac{1}{4} \left(\frac{2}{i\pi n} - \frac{2}{i\pi n} i^n - \frac{2}{i\pi n} (-1)^n + \frac{2}{i\pi n} (-i)^n - \frac{2}{i\pi n} (-i)^n + \frac{4}{\pi^2 n^2} (-i)^n + \frac{4}{\pi^2 n^2} \right) \\ &= \frac{1}{4} \left(\frac{2}{i\pi n} - \frac{2}{i\pi n} i^n + \frac{2}{i\pi n} (-1)^n + \frac{4}{\pi^2 n^2} (-i)^n + \frac{4}{\pi^2 n^2} \right) \end{aligned}$$

$$f(x) = \frac{5}{8} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} d_n e^{i\frac{\pi}{2}nx}$$

Problem 2: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.



$$d_0 = 0$$

$$d_n = \frac{-i}{2\pi} \int_0^{2\pi} f(x) \sin\left(\frac{n}{2}x\right) dx$$

$$= -\frac{i}{2\pi} \int_0^{\pi} \pi \sin\left(\frac{n}{2}x\right) dx$$

$$= -\frac{i}{2} \left(-\frac{2}{n} \cos\left(\frac{n}{2}x\right) \right) \Big|_0^{\pi}$$

$$= \frac{i}{n} \left(\cos\left(\frac{\pi}{2}n\right) - \cos(0) \right) = \frac{i}{n} \left(\cos\left(\frac{\pi}{2}n\right) - 1 \right)$$

$$f(x) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} d_n e^{i\frac{n}{2}x}$$

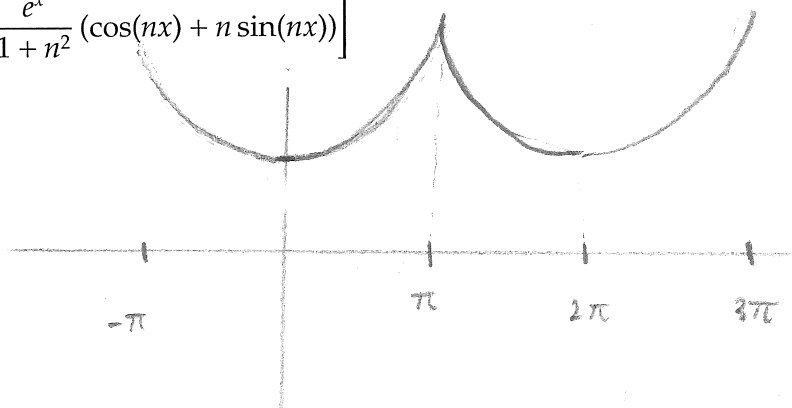
Problem 3: Consider the periodic function $f(x) = \begin{cases} e^{-x} & -\pi \leq x \leq 0 \\ e^x & 0 < x \leq \pi \end{cases}$ and $f(x) = f(x + 2\pi)$

(a) Plot $f(x)$ on the interval $[-\pi, 3\pi]$ (yes, I meant 3π)

(b) Find the Fourier Series of $f(x)$

[Hint: An anti-derivative of $e^x \cos(nx)$ is $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$]

(a)



Even

$L = \pi$

$$(b) \quad d_0 = \frac{1}{\pi} \int_0^{\pi} e^x dx = \frac{1}{\pi} e^x \Big|_0^{\pi} = \frac{1}{\pi} (e^{\pi} - 1)$$

$$d_n = \frac{1}{\pi} \int_0^{\pi} e^x \cos(nx) dx = \frac{1}{\pi} \left(\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx)) \right) \Big|_0^{\pi}$$

$$= \frac{1}{\pi} \left(\frac{e^{\pi}}{1+n^2} (-1) - \frac{1}{1+n^2} (1) \right)$$

$$= \frac{-e^{\pi} - 1}{\pi(1+n^2)}$$

$$f(x) = \frac{e^{\pi} - 1}{\pi} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{-e^{\pi} - 1}{\pi(1+n^2)} e^{inx}$$

Extra Credit: Show that $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$ is an anti-derivative of $e^x \cos(nx)$

Note: Do not just take the derivative of $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$

$$\int e^x \cos(nx) dx$$

$$u = \cos(nx) \quad v = e^x$$
$$du = -n \sin(nx) \quad dv = e^x dx$$

$$= e^x \cos(nx) + n \int e^x \sin(nx) dx$$

$$u = \sin(nx) \quad v = e^x$$
$$du = n \cos(nx) \quad dv = e^x dx$$

$$= e^x \cos(nx) + n \left[e^x \sin(nx) - n \int e^x \cos(nx) dx \right]$$

$$= e^x \cos(nx) + n e^x \sin(nx) - \underbrace{n^2 \int e^x \cos(nx) dx}_{**}$$

$$\Rightarrow \underbrace{\int e^x \cos(nx) dx}_{*} + \underbrace{n^2 \int e^x \cos(nx) dx}_{**} = e^x (\cos(nx) + n \sin(nx))$$

$$\Rightarrow (1+n^2) \int e^x \cos(nx) dx = e^x (\cos(nx) + n \sin(nx))$$

$$\Rightarrow \int e^x \cos(nx) dx = \frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$$